

Minimum mean brightness error contrast enhancement of color images using adaptive gamma correction with color preserving framework



Bhupendra Gupta*, Mayank Tiwari

Indian Institute of Information Technology, Design & Manufacturing Jabalpur, 482005 MP, India

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ABSTRACT

In this paper, we propose an efficient method to enhance contrast of digital images. Image contrast enhancement is a pre-processing step that improves efficiency of other image dealing applications such as computer vision and pattern recognition. The proposed method enhances contrast and brightness of given image using the gamma correction and weighted probability distribution of luminance pixels. Experiments results show that the proposed method is able to enhance contrast of all type of color images without much affecting its visual and color information.

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1. Introduction

Insufficient contrast in back-light images reduce the ability of observer in analyzing the image. In many cases insufficient contrast may produce difficulties in subsequent processing. This insufficient contrast may occur due to various reasons such as presence of clouds in the captured scene, lack of operator expertise and inadequacy of the image capture device, etc. In the available literature we are giving introduction of various methods that are used for image contrast enhancement. These methods can be broadly divided into two categories they are, direct enhancement [2–4] and indirect enhancement [5–7]. In direct enhancement methods, the image contrast can be directly defined by a specific contrast term [2–4]. However, most of these metrics cannot simultaneously gauge the contrast of simple and complex patterns in images which contain both [4].

Histogram equalization (HE) based techniques fall under the category of indirect enhancement methods. These techniques attempt to enhance image contrast by redistributing the probability density [1]. Using these techniques [9–15] the intensities can be better redistributed within the dynamic range. Histogram equalization techniques [9–15] are popular due to their simplicity

of implementation and computationally less expensiveness than other methods.

Although the HE based methods are widely used, in histogram equalization we do not have any mechanism to control the enhancement level, due to this sometimes output image may have over enhanced regions. Also HE could not effectively work, when the input image contains regions that are significantly darker or brighter than other parts of the image. Another problem with HE is, in theory it is clearly shown that mean brightness of histogram equalize image is always middle gray level; regardless the mean brightness of input image. This problem is known as the ‘mean-shift’ problem. In [8], K. Zuiderveld proposed a widely used method known as ‘contrast limited adaptive histogram equalization’ (CLAHE). The CLAHE method was able to reduce the over enhancement of contrast in the processed image but it was not capable in reducing the mean brightness change in the processed image. To solve the ‘mean-shift’ problem, Yeong-Taeg in [9] proposed ‘brightness preserving bi-histogram equalization’ (BBHE) where the image X is divided into two sub-images X_L and X_U based on mean X_M of brightness of input image and then equalize these two sub-histograms separately. Later Zhang in [10] proposed ‘equal area dualistic sub-image histogram equalization’ (DSIHE), though this approach also separates an input image into two sub-sections, the only difference between BBHE and DSIHE is, in later method the separation is based on the median value.

An extension of BBHE has been proposed by Chen and Ramli in [11], known as ‘minimum mean brightness error bi-histogram

* Corresponding author. Tel.: +91 9425155354.

E-mail addresses: gupta.bhupendra@gmail.com (B. Gupta), mayanktiwarigigs@gmail.com (M. Tiwari).

URL: <http://https://www.bhupendragupta.com> (B. Gupta).

equalization' (MMBEBHE). This method provides maximum brightness preservation. The MMBEBHE proposes to perform the separation based on the threshold level, which would yield minimum absolute mean brightness error [11]. The MMBEBHE is a useful tool to control the brightness difference between input and processed image. In [12], Chen and Ramli proposed another interesting method called 'recursive mean-separate histogram equalization' (RMSHE), here authors suggested recursive division of histograms, based on the local mean. In each recursive step existing sub-histogram is divided into two sub-histograms. After r th recursion the number of sub-histogram is 2^r , where number of recursion depends on choice of user. Also authors proved mathematically that as r increases, the mean brightness of processed image approaches towards the mean brightness of input image. Sim et al. in [13], improved DSIHE into 'recursive sub-image histogram equalization' based contrast enhancement (RSIHE), by introducing recursive segmentation in the similar manner as Chen and Ramil proposed in [12], this method is similar to RMSHE but it uses median values instead of mean values to divide histogram into sub-histograms.

Recently, S.C. Huang et al. [16], propose 'efficient contrast enhancement using adaptive gamma correction with weighting distribution' (AGCWD). This method is an automatic transformation technique that improves the brightness of dimmed images via the gamma correction and probability distribution of luminance pixels; also this method produces enhanced images of comparable or higher quality than those produced by other methods [16].

The organization of this work is as follows. After introducing about research background in Section 1, Section 2 describes the AGCWD method. Section 3 explains the proposed method in details. For evaluation of proposed method experimental results are shown in Section 4. Finally Section 5 summarizes and concludes the proposed work.

2. The AGCWD method

As mentioned earlier in [16], S.C. Huang et al. proposes 'efficient contrast enhancement using adaptive gamma correction with weighting distribution', (AGCWD). The AGCWD method can progressively increase the low intensity and avoid the significant decrement of the high intensity. The AGCWD method applies the weighting distribution (WD) function to slightly modify the statistical histogram and lessen the generation of adverse effects. The WD function is formulated as:

$$pdf_w(l) = pdf_{\max} \left(\frac{pdf(l) - pdf_{\min}}{pdf_{\max} - pdf_{\min}} \right)^{\alpha}, \quad (1)$$

where α is the adjusted parameter, $pdf_{\max}$ is the maximum pdf of the statistical histogram, and $pdf_{\min}$ is the minimum pdf . Based on Eq. (1) the gamma parameter is given as:

$$\gamma = 1 - cdf(l), \quad (2)$$

where cdf is resultant cumulative distribution function calculated after normalizing pdf_w . The adaptive gamma correction (AGC) used by S.C. Huang et al. in [16] is given as:

$$T(l) = (L-1) \left(\frac{l}{L-1} \right)^{\gamma} = (L-1) \left(\frac{l}{L-1} \right)^{1-cdf(l)}, \quad (3)$$

where $(L-1)$ is the maximum intensity value of the image. It is clear that slight increase in $cdf(l)$ will cause drastic change in intensity value l of the processed image.

3. The proposed method

The resultant cdf curve of AGCWD [16] method is shown in Fig. 1.

It is clear from Fig. 1 that for $\alpha = 1.2$ and $\alpha = 0.8$ the resultant transformation curve is not increasing smoothly. To prove our this claim (in *image*₁ for transformation curve $\alpha = 1.2$) we have marked two intervals on x -axis (x_1, x_2); (x_3, x_4) and their corresponding intervals on y -axis (y_1, y_2); (y_3, y_4). The small interval (x_1, x_2) is transformed to sufficiently large interval (y_1, y_2) and in this case sufficient weights will be assigned to Eq. (3). However on the other hand due to non-smooth nature of the curve the large interval (x_3, x_4) is transformed to a small interval (y_3, y_4) and hence weights for a large interval will be not sufficiently large. This may also lead to loss of information in processed image.

To solve this problem and to make the resultant transformation curve to be smoothly increasing with sufficient color preservation; we proposed a new method called 'adaptive gamma correction with color preserving framework' (AGCCPF). The proposed method uses two-steps processing; initially it enhances contrast and brightness of given image using modified probability distribution of luminance pixels followed by the gamma correction. In second step it uses a color preserving framework for color restoration.

The objective of all histogram based contrast enhancement methods is to transform the histogram of an input image closer

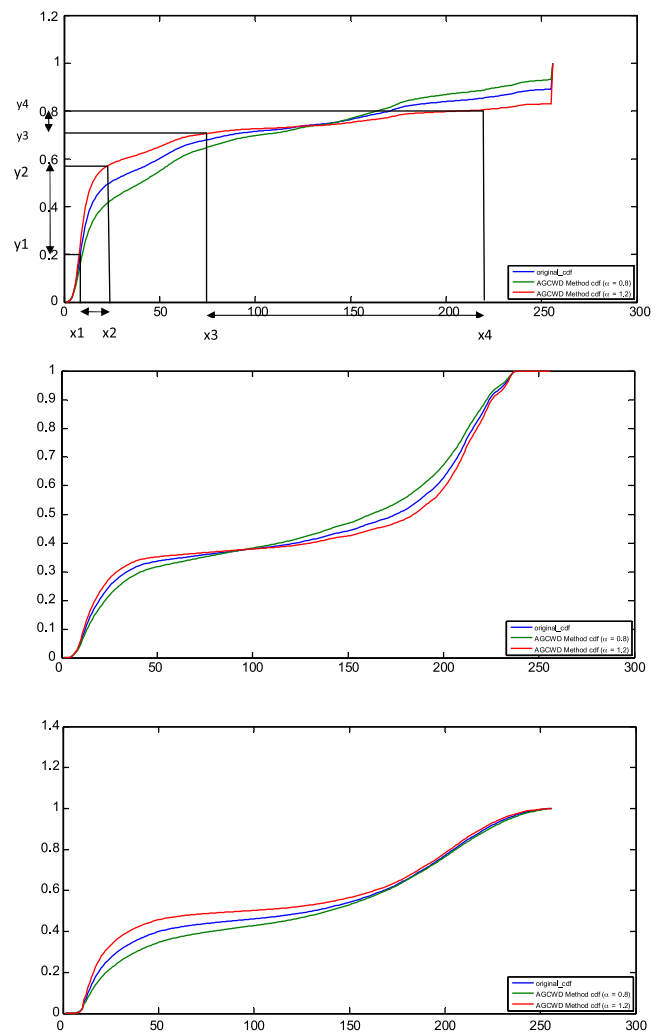


Fig. 1. Resultant transformation curve of AGCWD method, after applying weighting distribution function on *image*₁, *image*₂ and *image*₃. Here x -axis shows gray level of image and y -axis shows corresponding cumulative distribution function at each gray level.

to uniform histogram, and also the objective is to maximize the image contrast by transforming the gray level distribution of input image to a gray level distribution which is closer to uniform gray level distribution.

Here, we use a histogram modification function for contrast enhancement. This function has the ability to control the level of contrast enhancement in the processed image.

In histogram equalization we consider an image as a 2 dimensional array of gray levels. Let the (i, j) element of this array is $X(i; j)$ be the intensity of (i, j) pixel of the image, where $X(i; j)$ is from the L discrete gray levels denoted by $\{X_0, X_1, \dots, X_{L-1}\}$. Let H be the histogram of the input image and let H_u be the uniform histogram. The main idea is to obtain a modified histogram $\hat{H}$ from the input histogram H such that the difference between $\hat{H}$ and H_u is sufficiently small, keeping the $\hat{H}$ closer to H . Now it is an optimization problem and can be formalized as:

$$\hat{H} = \theta H + (1 - \theta)H_u, \quad (4)$$

where $0 \leq \theta \leq 1$.

In Fig. 2, we are showing mapping function for different values of θ . This mapping function is nothing but the cumulative distribution function of resultant modified histogram. For $\theta = 0.5$ the mapping curve increases smoothly to the maximum value and this causes no over enhancement in the processed image. Now based on Eq. (4) the corresponding normalized probability mass function (PMF) can be given as:

$$pmf(l) = \frac{\hat{H}(l)}{\sum \hat{H}}, \quad (5)$$

where $\sum \hat{H} = \sum_{k=0}^{L-1} \hat{H}(k)$. Now based on Eq. (5) the corresponding cdf is given as:

$$cdf(l) = \sum_{k=0}^l pmf(k). \quad (6)$$

The proposed method directly utilizes the cdf (which is generated after histogram modification function). This cdf is applied in a normalized gamma function which modifies the transformation curve. After calculation of cdf using Eq. (6) we apply adaptive gamma correction using Eqs. (2) and (3).

By the histogram modification function we can adjust the cdf of input image so that the resultant cdf is smoothly increasing and hence no over enhancement will occur. However when we deal with color images then our aim is not only to enhance their contrast but also to preserve their color information up to some extent. For preservation of color information in processed image, we are using

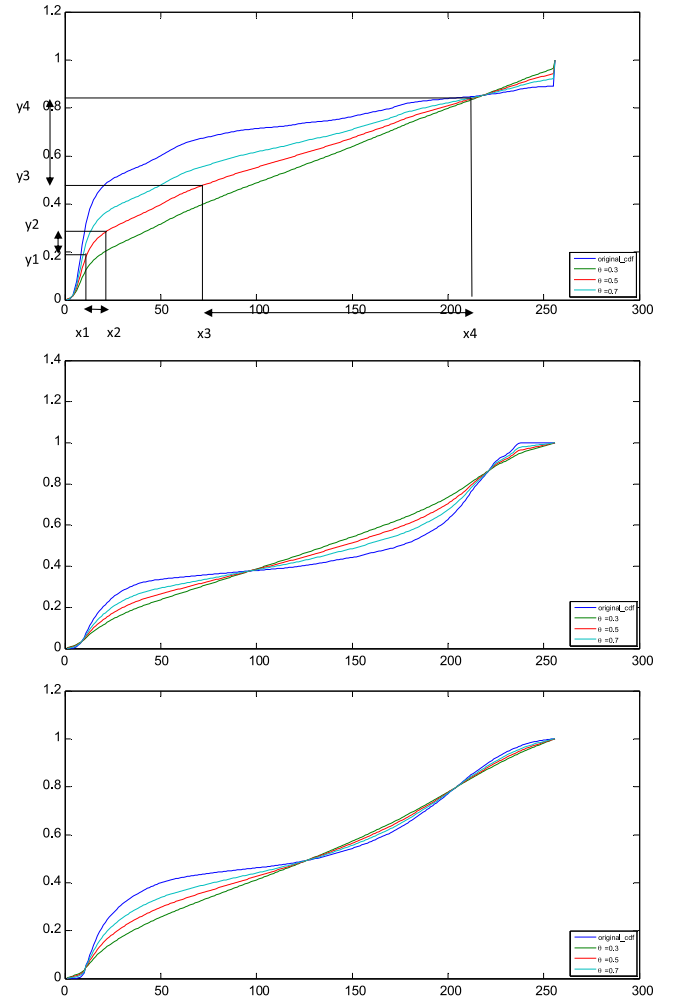


Fig. 2. Resultant transformation curve after applying histogram modification function. Here the transformation curves are shown for $image_1$, $image_2$ and $image_3$. Here x-axis shows gray level of image and y-axis shows corresponding cumulative distribution function at each gray level.

the color information of input image and applying it in processed image. Now we can restore color information in processed image by the following:

$$PRC_{IMG} = \delta(INT_{IMG}) + (1 - \delta)(INP_{IMG}), \quad (7)$$

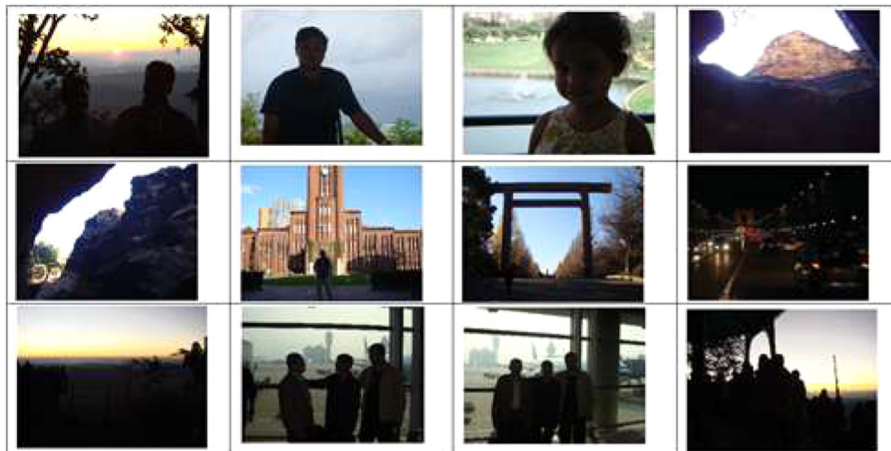


Fig. 3. Set of images used in the proposed work, images are given names as $image_1$, $image_2$, up to $image_{12}$.

where INT_{IMG} is image obtained after applying Eq. (3), INP_{IMG} is input image and PRC_{IMG} is corresponding processed image. Parameter $0 \leq \delta \leq 1$. It is clear that for $\delta = 0$ $PRC_{IMG} = INP_{IMG}$ and in this case there would be maximum color preservation without any contrast enhancement in the PRC_{IMG} . For $\delta = 1$ $PRC_{IMG} = INT_{IMG}$ and in this case there would be minimum color preservation with contrast enhancement.

In the proposed work we are working with color images using the RGB color model. The RGB color model is an additive color model in which red, green, and blue light are added together in various ways to reproduce a broad array of colors. The name of the model comes from the initials of the three additive primary colors, red, green, and blue. In implementation part of proposed algorithm, we are applying the algorithm on each color channel of given color image, after that we are combining all three channels to form the processed image.

4. Experimental results

In this section, we demonstrate performance of the proposed method in comparison with some existing HE based contrast enhancement methods, like HE, RMSHE, RSIHE, RSWHE and AGCWD. For RMSHE, RSIHE and RSWHE we are using recursion level $r=1$. As test images we are taking images having poor

illumination due to back-light effect. Generally change in illumination can be due to many factors, such as location of light source, cloud cover, intensity of sunshine, etc. Fig. 3 shows set of images, used by us in the proposed work.

Fig. 4 shows enhancement results of various methods for $image_2$. The $image_2$ contains scene with an object and clear sky background. The HE method produces over enhancement and noise enhancement too; in the processed image (b). Results of other methods such as RMSHE and RSIHE are able to enhance the contrast of the image at the cost of natural appearance's loss. The CLAHE method is producing better results but this method is not able to maintain the mean brightness in the processed image. The AGCWD method produces good results but in this case color-information of the sky in the background is washed out. On the other hand the AGCCPF method is able to enhance contrast of given image without much affecting its color-information and natural appearance.

Table 1 shows results of Absolute Mean Brightness Error (AMBE). AMBE is used to calculate difference in mean brightness between two images. Mathematical expression to calculate AMBE between two images is given as:

$$AMBE = |X_M - Y_M|, \quad (8)$$

where X_M and Y_M are mean brightness of input and processed image respectively.

Based on results of Table 1, we observe that the proposed method has least values in all 12 images as compare to other methods. Further if we look at last row of Table 1, which shows average

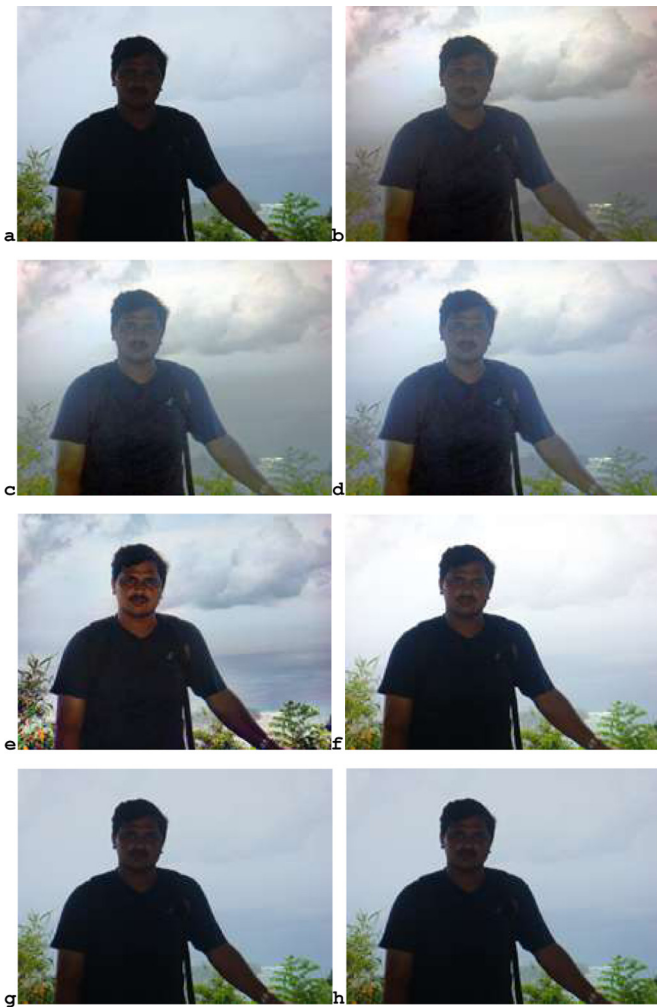


Fig. 4. Enhancement results of various methods for $image_2$. Here (a) input image, (b) result of HE, (c) result of RMSHE ($r=1$), (d) result of RSIHE ($r=1$), (e) result of CLAHE ($AMBE=6.514$), (f) result of AGCWD ($\alpha=0.8$), (g) proposed method ($\theta=0.4$ and $\delta=0.4$) ($AMBE=0.011$), (h) proposed method ($\theta=0.25$ and $\delta=0.4$).

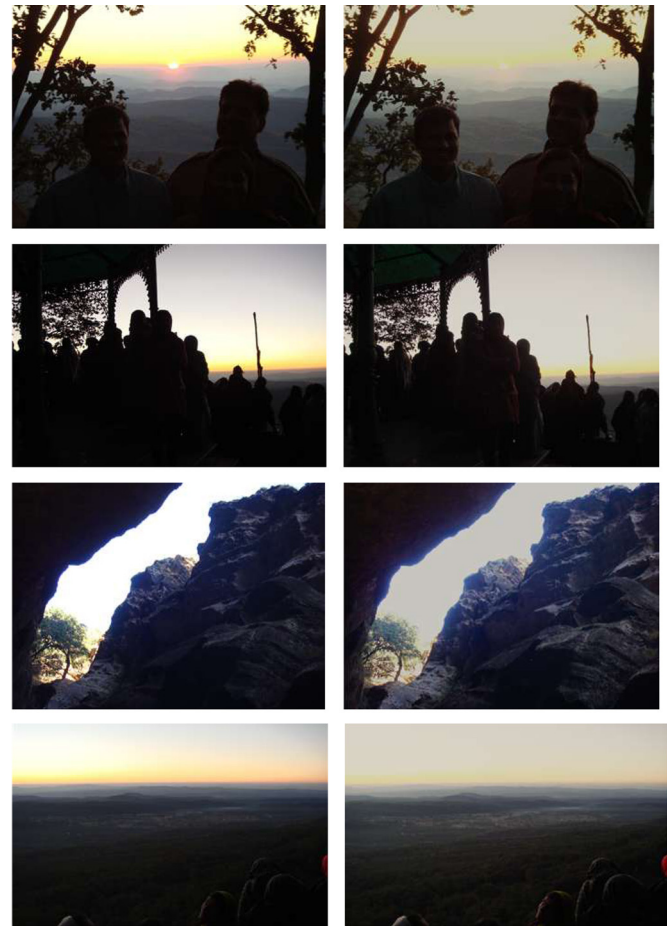


Fig. 5. Enhancement results of proposed method for $image_1$, $image_{12}$, $image_5$ and $image_9$. Here first column shows the input images second column shows corresponding processed images. For these 4 images we have considered $\theta=0.4$ and $\delta=0.4$.

Table 1
Absolute Mean Brightness Error (AMBE).

Image name	HE	RMSHE	RSIHE	CLAHE	AGCWD	AGCCPF
<i>image</i> ₁	56.834	8.750	4.380	17.055	33.376	0.024
<i>image</i> ₂	35.338	1.784	8.324	6.514	29.170	0.011
<i>image</i> ₃	58.271	7.797	7.088	5.656	22.077	0.036
<i>image</i> ₄	40.030	5.952	1.411	13.893	32.549	0.070
<i>image</i> ₅	63.714	4.504	14.058	21.957	40.720	0.001
<i>image</i> ₆	2.382	5.522	3.010	7.077	33.263	0.026
<i>image</i> ₇	42.255	7.026	1.705	20.659	36.037	0.024
<i>image</i> ₈	119.163	24.613	50.735	22.932	48.232	0.029
<i>image</i> ₉	3.278	10.831	18.494	4.334	27.806	0.006
<i>image</i> ₁₀	50.577	8.043	3.217	17.511	34.718	0.001
<i>image</i> ₁₁	37.607	3.070	4.218	16.417	31.385	0.005
<i>image</i> ₁₂	13.849	10.385	16.100	0.298	30.044	0.018
Average	43.608	8.190	11.062	12.858	33.281	0.021

results of AMBE then we find that the proposed method has least average AMBE values among other methods. This shows that our method is able to maintain mean brightness of input image in the processed image.

Table 2 shows results of Peak Signal to Noise Ratio matrix. Peak Signal to Noise Ratio (PSNR) is widely used matrix which is used to measure contrast enhancement [13,14,17,18]. Greater PSNR values indicate better the image quality [14]. To calculate PSNR between two images (each image is having L discrete gray levels in the range $\{X_0, X_1, \dots, X_{L-1}\}$), mathematical expression is given as:

$$PSNR = 10 \log_{10} \frac{(L-1)^2}{MSE}, \quad (9)$$

where MSE is Mean Square Error and it is defined as:

$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N |X(i,j) - Y(i,j)|^2, \quad (10)$$

where $X(i,j)$ and $Y(i,j)$ denotes intensity at location (i,j) in the input and processed image respectively.

Table 2 proves that proposed method enhances more contrast than HE, RMSHE, RSIHE, CLAHE and AGCWD (as higher PSNR value indicates better contrast enhancement [13,14,17,18]). On the basis



Fig. 6. Enhancement results of proposed method for *image*₄, *image*₆, *image*₇ and *image*₈. Here first column shows the input images second column shows corresponding processed images. For these 4 images we have considered $\theta=0.4$ and $\delta=0.4$.



Fig. 7. Enhancement results of proposed method for *image*₂, *image*₁₀, *image*₁₁, and *image*₃. Here first column shows the input images second column shows corresponding processed images. For these 4 images we have considered $\theta=0.4$ and $\delta=0.4$.

Table 2
Peak Signal to Noise Ratio (PSNR).

Image name	HE	RMSHE	RSIHE	CLAHE	AGCWD	AGCCPF
<i>image</i> ₁	11.43	21.25	25.25	19.46	14.92	23.65
<i>image</i> ₂	13.65	17.67	19.53	21.40	16.53	24.62
<i>image</i> ₃	10.31	15.80	18.88	22.01	18.90	26.88
<i>image</i> ₄	13.49	21.68	25.36	19.96	15.21	21.83
<i>image</i> ₅	9.98	23.19	17.66	18.69	13.18	23.18
<i>image</i> ₆	23.04	21.68	22.63	17.47	16.29	23.42
<i>image</i> ₇	13.66	20.33	22.73	17.81	15.20	26.05
<i>image</i> ₈	5.78	13.95	9.90	18.53	11.45	31.29
<i>image</i> ₉	18.03	18.28	18.19	21.73	18.26	27.96
<i>image</i> ₁₀	12.71	22.28	26.94	19.27	15.46	26.70
<i>image</i> ₁₁	13.94	20.31	23.16	18.17	15.78	22.80
<i>image</i> ₁₂	19.39	20.15	19.18	16.99	17.37	26.14
Average	13.79	19.71	20.79	19.29	15.71	25.38

of results of [Tables 1 and 2](#) we can conclude that the AGCCPF is a good method for contrast enhancement with maximum brightness preservation and color preservation.

In the following [Figs. 5–7](#) we are showing results of proposed method on all 12 images.

5. Conclusion

The AGCCPF method provides optimum contrast enhancement while preserving the brightness of given color image and hence this method is suitable for all types of color images. Experimental results show that AMBE of the proposed method is less in comparison of other methods. Also PSNR of the proposed method is better from HE, RMSHE, RSIHE, CLAHE and AGCWD. On the basis of analysis of these two metrics, it is clear that AGCCPF preserves the input image brightness more accurately and gives processed image with good contrast enhancement.

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